

Acing the Math Section of the Digital SAT

Tips, Tricks and Warnings

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Overview of Test Structure

The Math section has 2 modules: Module 1 and Module 2.

Each module is **35 minutes long** with **22 questions each**, so you have about 95 seconds per question.

Module 1 is almost the same for everyone, and serves as a starting point. It should be medium difficulty.

If you perform well on Module 1, you get a harder Module 2. If you perform poorly on Module 1, you get an easier Module 2. So, by looking at the difficulty of Module 2, you can have an idea of your performance in Module 1.

General Advice: Exam Approach and Preparation

You should already be using this ([Khan Academy](#)) on a weekly basis to prepare. Here is a sample page from the Digital SAT Math section:

The screenshot shows the Khan Academy interface for the Digital SAT Math section. The top navigation bar includes 'Courses', 'Search', the Khan Academy logo, 'Donate', and a user profile 'tonyrotich2580'. On the left, a sidebar lists units from Unit 1 to Unit 8. The main content area displays Unit 1, Unit 2, and Unit 3. Unit 1 is titled 'Unit 1: About the digital SAT' and includes topics like 'What is the new digital SAT?' and 'Preparing for the SAT'. Unit 2 is titled 'Unit 2: Foundations: Algebra' and shows a progress bar for '580/800 Mastery points'. It lists topics such as 'Solving linear equations and inequalities: foundations', 'Linear equation word problems: foundations', 'Linear relationship word problems: foundations', 'Graphs of linear equations and functions: foundations', 'Solving systems of linear equations: foundations', 'Systems of linear equations word problems: foundations', 'Linear inequality word problems: foundations', and 'Graphs of linear systems and inequalities: foundations'. Unit 3 is titled 'Unit 3: Foundations: Problem solving and data analysis' and shows a progress bar for '510/1000 Mastery points'. It lists topics such as 'Ratios, rates, and proportions: foundations', 'Unit conversion: foundations', 'Percentages: foundations', 'Center, spread, and shape of distributions: foundations', 'Data representations: foundations', 'Scatterplots: foundations', 'Linear and exponential growth: foundations', 'Probability and relative frequency: foundations', 'Data inferences: foundations', and 'Evaluating statistical claims: foundations'.

- One piece of advice is that you should be doing the topical practice questions frequently. Frequently does not mean once a week. If you are genuinely dedicated, you should do practice every day or every 2 or 3 days.
- Another is that you should do the Unit Tests found at the bottom of each Unit. This will give you a clear picture as to **WHAT YOU KNOW** and **WHAT YOU DON'T KNOW**. Khan

Academy is great in that it shows you YOUR STRENGTHS and YOUR WEAKNESSES, and gives you appropriate RECOMMENDATIONS to improve, so use these features to your benefit. Also, if you fail a question, find out WHY YOU FAILED IT and CORRECT YOUR MISTAKES.

- Don't focus too much on ONE PARTICULAR AREA! For example, don't spend a disproportionate amount of time practicing Algebra questions and forgetting Geometry questions. Remember, the exam will test ALL THESE AREAS.
- Try to avoid MENTAL MATH calculations. For example, in the following question where you need to find x:

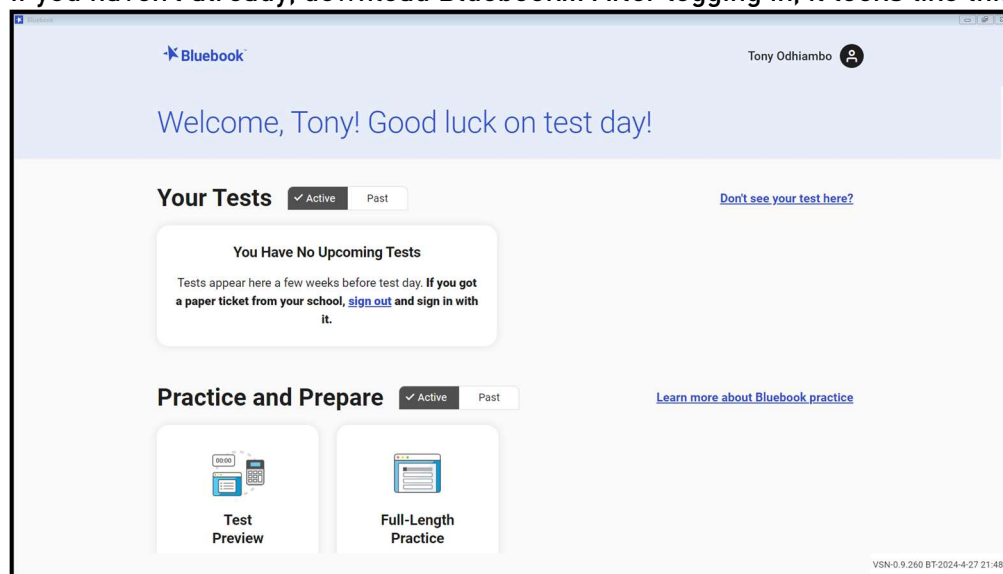
$$\frac{3}{4}x + 4 = 20$$

- There is no need to mentally subtract $20 - 4$ to get 16 and thereafter multiply 16 by $4/3$ to get $64/3$. You are likely to make mistakes, particularly when you are under pressure. Maybe you might forget and do the following instead in your head:

$$\begin{aligned}\frac{3}{4}x &= 20 - 4 = 16 \\ x &= 16 \times \frac{4}{3} = 4 \times 3 = 12\end{aligned}$$

just because you noticed 16 is divisible by 4. The thing is, you will probably see 12 as an answer choice and you will pick it. The SAT makers are familiar with such student errors and intentionally include such answer choices.

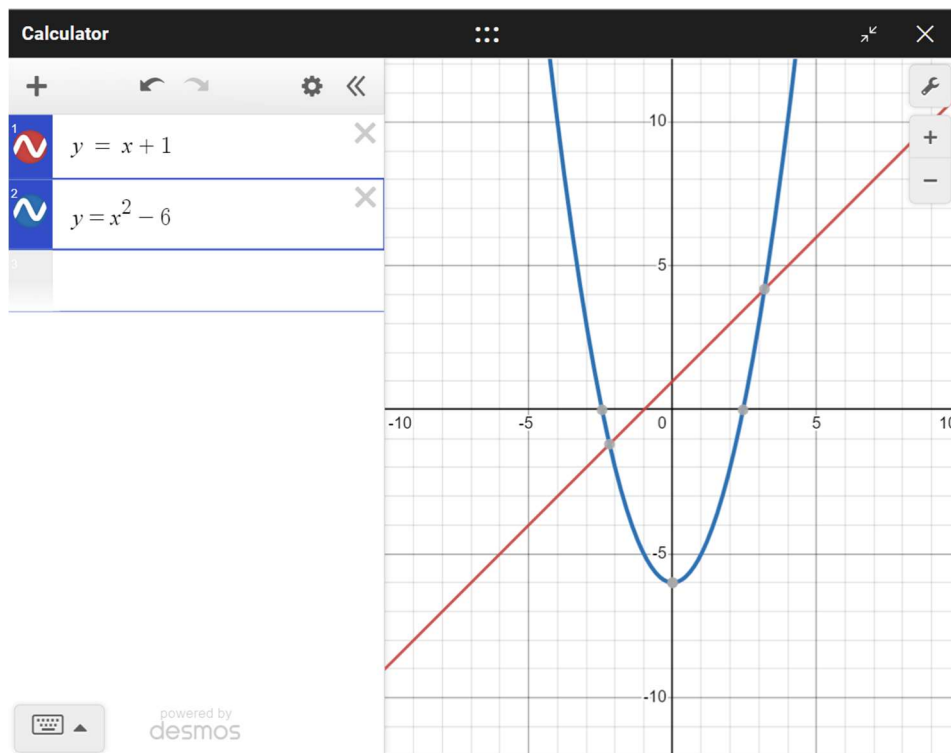
- THEREFORE, USE YOUR CALCULATOR and WRITE DOWN YOUR WORK. I cannot emphasize this enough. You cannot verify mental math, but you CAN VERIFY written work.
- In the SAT, you will not carry paper for rough work. Instead you will be given 3 BLANK SHEETS OF PAPER, or 6 pages, by the test centre. USE THEM WELL! Bonus points if you can keep your work neatly organized, since in such a case, you will be able to spot silly errors you might have made. Don't worry, these papers will not be marked.
- If you haven't already, download Bluebook!!! After logging in, it looks like this:



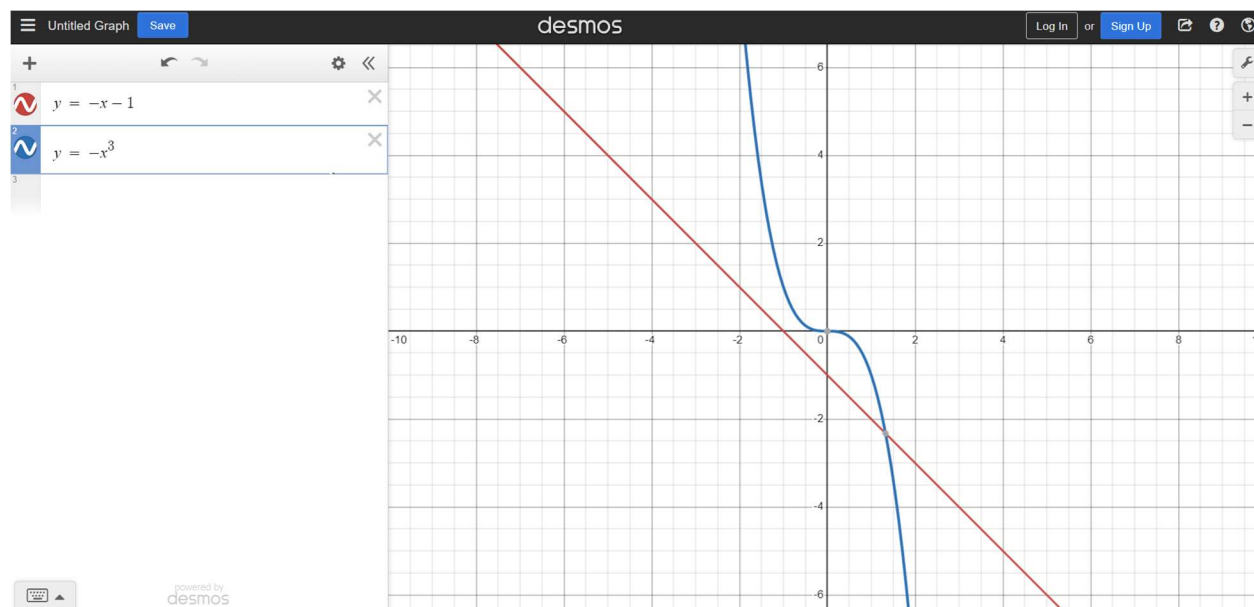
Calculator Usage and Tips

- In the math section, The SAT allows you to use a calculator the entire exam. Yes, THE ENTIRE EXAM. However, this alone will not help you. You need to be able to USE THE CALCULATOR, QUICKLY and ACCURATELY. The SAT is all about speed and precision.
- If you decide to use the handheld calculator, learn how to TYPE NUMBERS ACCURATELY, and USE THE CORRECT OPERATIONS. For example, do not mistakenly use the Subtraction symbol instead of the Division symbol.
- If you are typing a number with many repeated digits, PLEASE CONFIRM THAT YOU HAVE THE CORRECT NUMBER OF DIGITS. For example, when typing 100000 you might hurriedly count and see that there are 6 zeros. This mistake carries over and you end up with an incorrect answer.
- In short, with the handheld calculator, BE PRECISE with every calculation. REVIEW THE CALCULATION HISTORY if you need to.
- Whether or not you decide to use the handheld calculator, you should definitely use the GRAPHING CALCULATOR in Bluebook. The same can be accessed on a website called Desmos.

Bluebook Graphing Calculator:



Desmos:



As you can see, you can plot multiple graphs at once, in DIFFERENT COLOURS so you don't confuse them.

Please, PLEASE, familiarize yourself with this platform. It will be a lifesaver in the exam. In Bluebook, click on the Calculator icon on the Top Right if you wish to open up the Calculator.

Section 2: Math

Directions ▾

0:00

Hide

Calculator

Reference

More

4

Mark for Review

abc

Figure A and figure B are both regular polygons. The sum of the perimeter of figure A and the perimeter of figure B is 63 inches. The equation $3x + 6y = 63$ represents this situation, where x is the number of sides of figure A and y is the number of sides of figure B. Which statement is the best interpretation of 6 in this context?

(A)

 Each side of figure B has a length of 6 inches.

(B)

 The number of sides of figure B is 6.

(C)

 Each side of figure A has a length of 6 inches.

(D)

 The number of sides of figure A is 6.

What do I mean by “lifesaver”? Well, here is a long set of questions that can be solved using Desmos. By the end, you should be able to appreciate how useful the calculator can be. You don't need Desmos for every question, but I would like to show that you can use the calculator for almost every question, if you so wish to.

I have taken questions from SAT Practice papers released by College Board.

1

What is 10% of 470?

- A) 37
- B) 47
- C) 423
- D) 460

Just type the expression in Desmos.



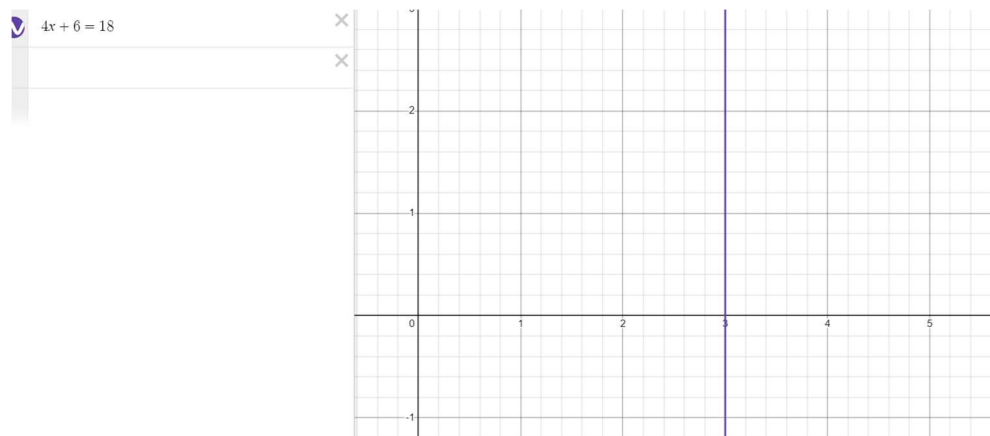
2

$$4x + 6 = 18$$

Which equation has the same solution as the given equation?

- A) $4x = 108$
- B) $4x = 24$
- C) $4x = 12$
- D) $4x = 3$

Same thing. Just type the expression in Desmos.



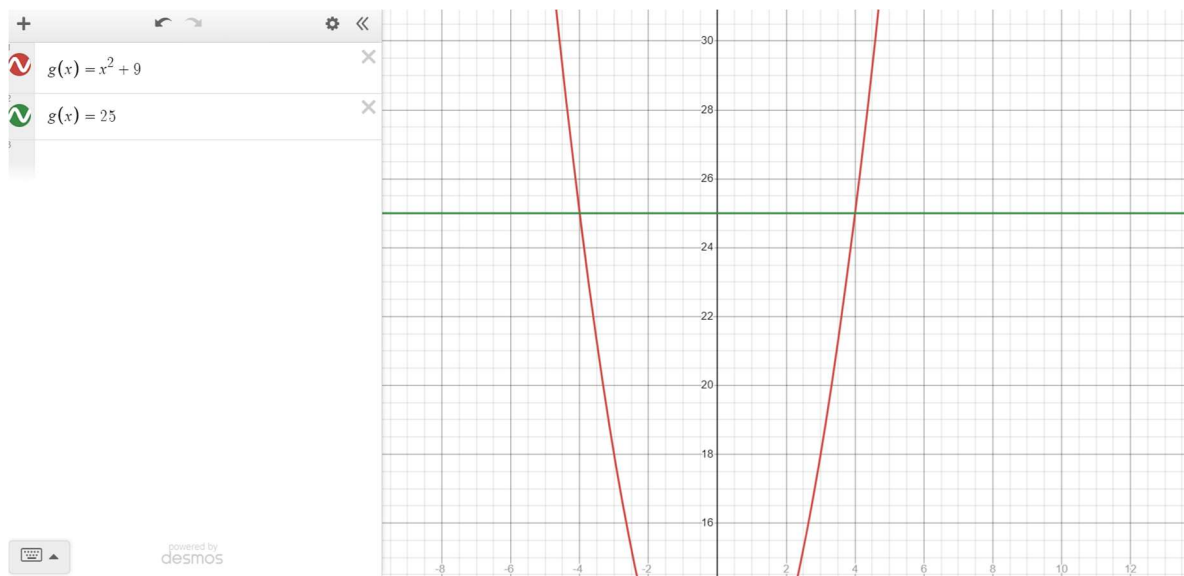
You can see clearly that $x = 3$ (DON'T FALL FOR THE TRAP AND PICK CHOICE D), and that should mean that $4x = 12$

4

The function g is defined by $g(x) = x^2 + 9$. For which value of x is $g(x) = 25$?

- A) 4
- B) 5
- C) 9
- D) 13

Type it into Desmos. Exactly as it is.



If you read the axes, you will see that $x = 4$ and $x = -4$ are solutions.

6

A printer produces posters at a constant rate of 42 posters per minute. At what rate, in posters per hour, does the printer produce the posters?

Even for such questions, you can just do all your calculations in-app.

The image shows a calculator interface. The input field contains the expression $42 \cdot \frac{60}{1}$. The result field shows the answer $= 2520$.

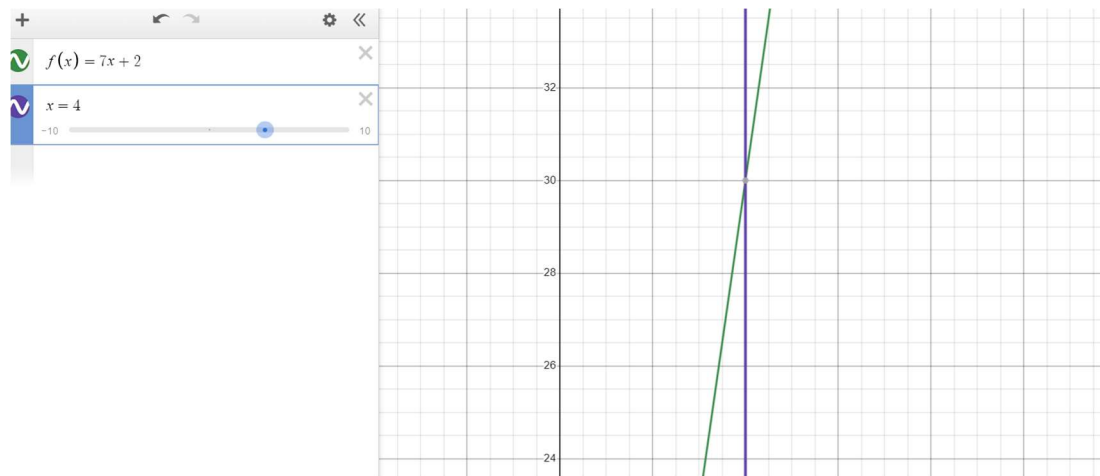
Moreover, in such questions, take a moment to think about the question, and the answer. 42 posters are produced per minute. PER MINUTE. One minute, 42, next minute, another 42. After 60 minutes, there should be 2520 posters produced.

DON'T EVER DO MATH QUESTIONS WITHOUT GIVING THEM A THOUGHT

7

The function f is defined by the equation $f(x) = 7x + 2$. What is the value of $f(x)$ when $x = 4$?

What should you do here? Simple, just type in every equation you see.



We can see that $f(x) = 30$

You can also see a sliding bar below $x = 4$. This means that you can move the line freely from one side to another and observe, if you need to.

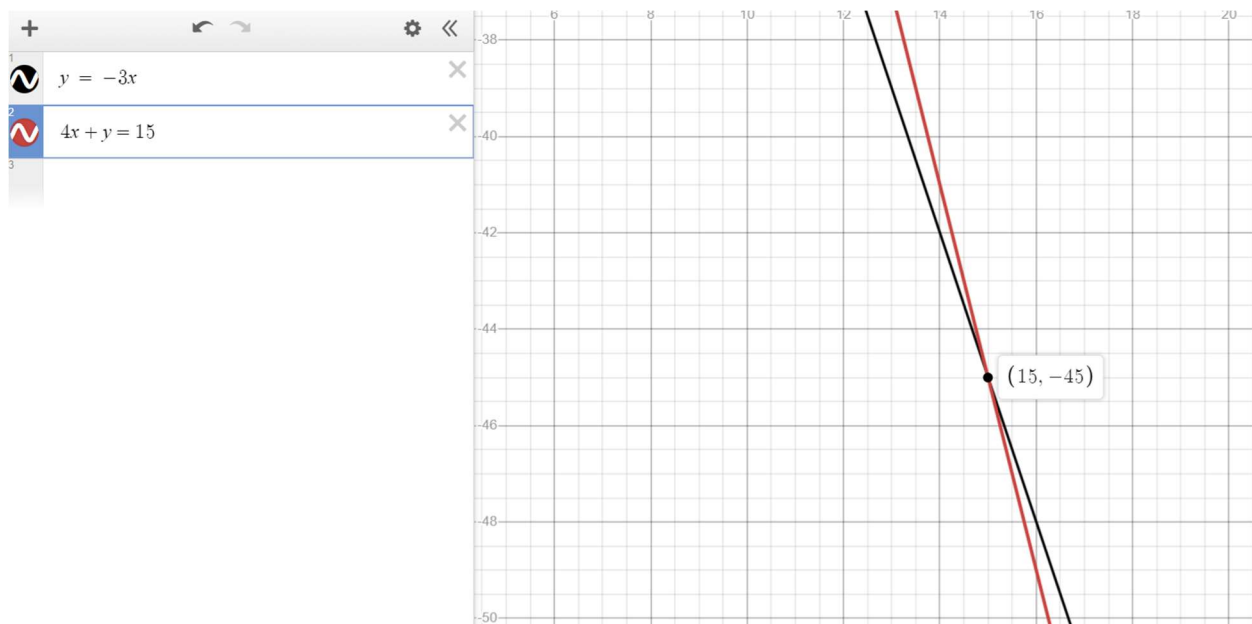
10

$$\begin{aligned}y &= -3x \\ 4x + y &= 15\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

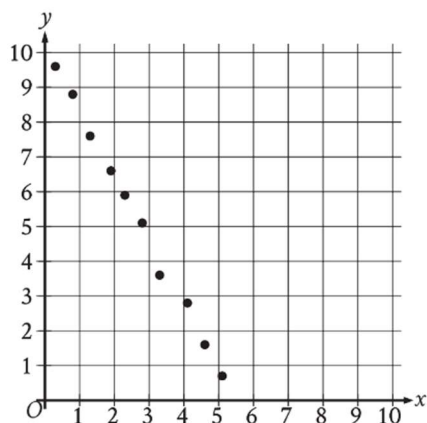
- A) 1
- B) 5
- C) 15
- D) 45

In such questions, you should always use Desmos.



Let's keep going.

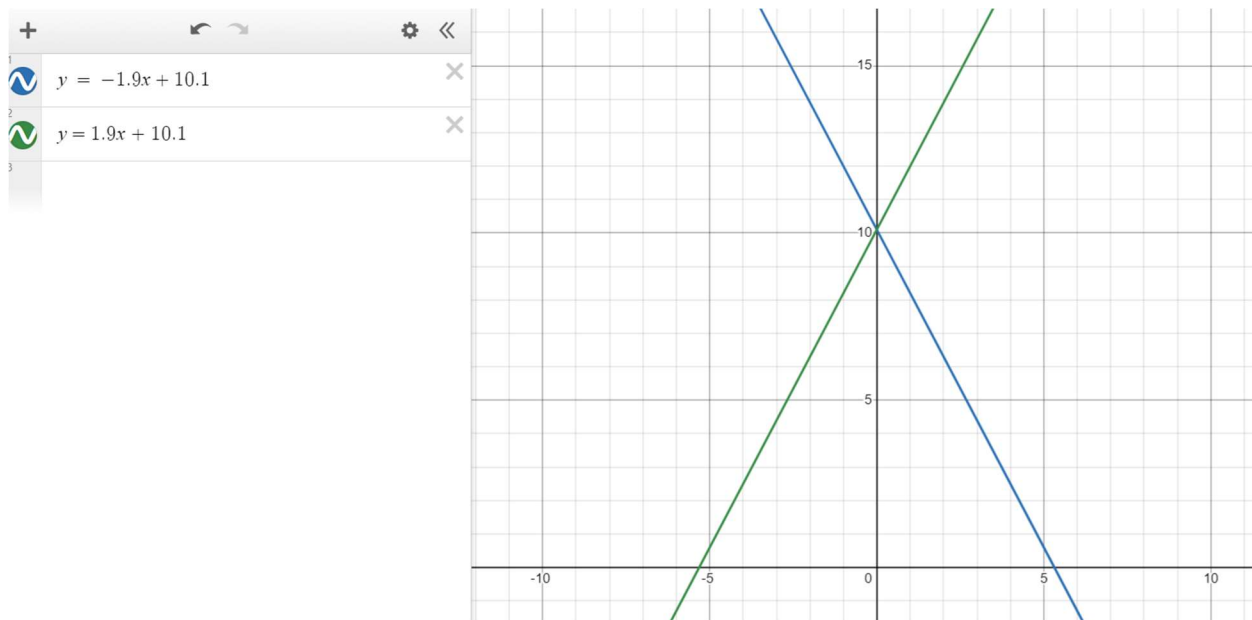
11



Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

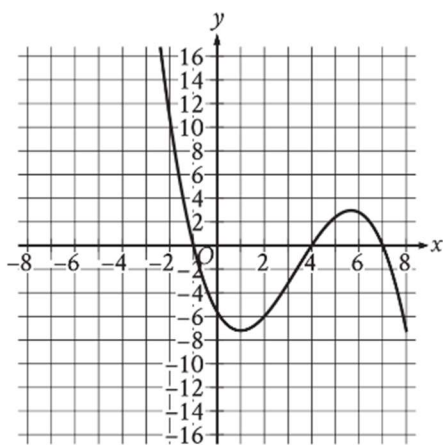
- A) $y = -1.9x - 10.1$
- B) $y = -1.9x + 10.1$
- C) $y = 1.9x - 10.1$
- D) $y = 1.9x + 10.1$

Although it might not seem as obvious as the other ones, the idea is simple. Type the 4 equations one by one and see which one has that shape. If you want to be quicker, use elimination and eliminate choices A and C because they have a negative y-intercept.



With this method. The answer should be very clear.

12



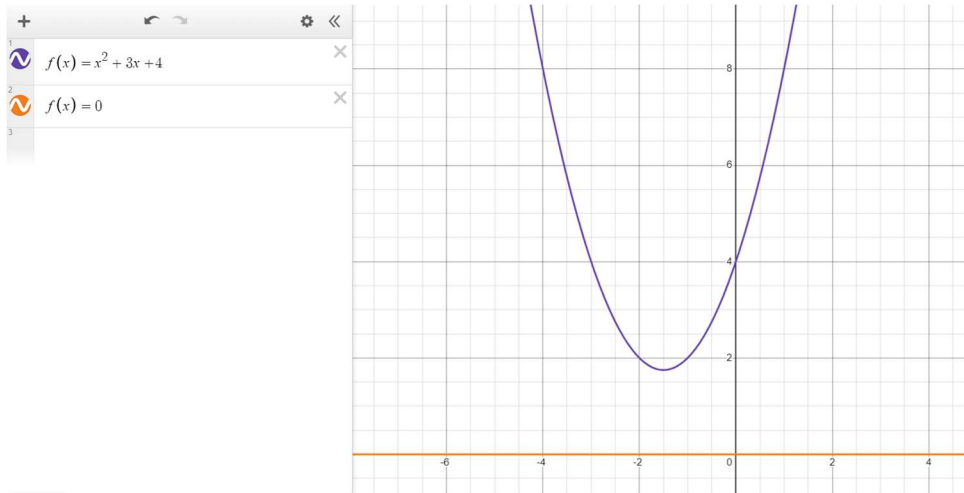
The graph of $y = f(x)$ is shown, where the function f is defined by $f(x) = ax^3 + bx^2 + cx + d$ and a , b , c , and d are constants. For how many values of x does $f(x) = 0$?

- A) One
- B) Two
- C) Three
- D) Four

For such a question, it should be obvious that the answer is 3. However, if the question asked this instead:

$$f(x) = x^2 + 3x + 4$$

And no graph was provided, you may then use Desmos.



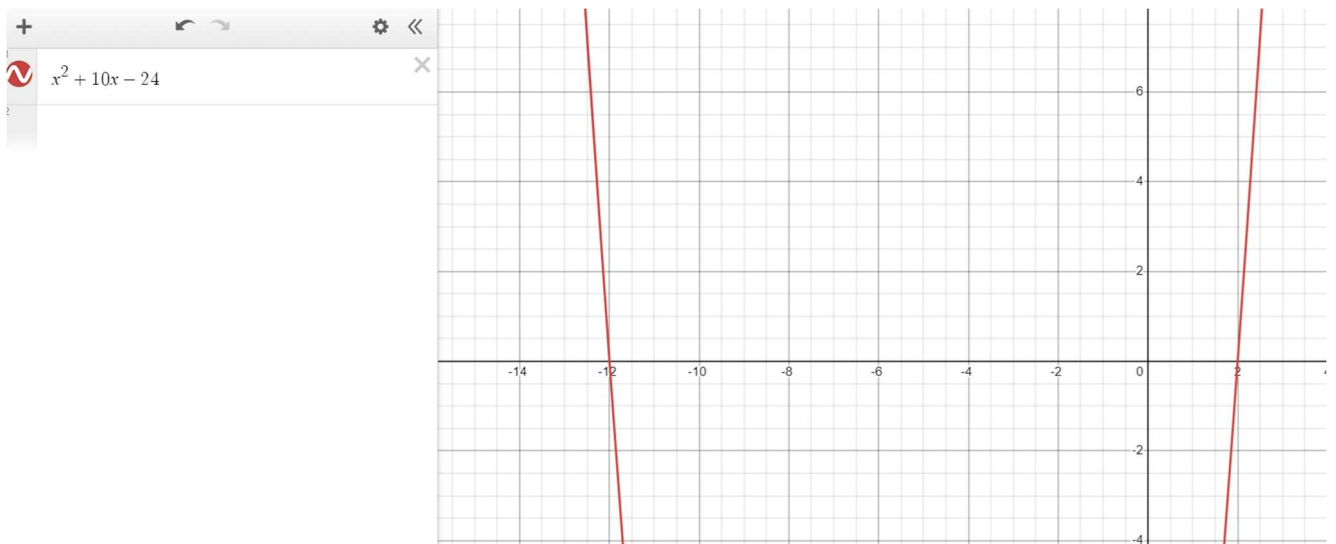
14

$$z^2 + 10z - 24 = 0$$

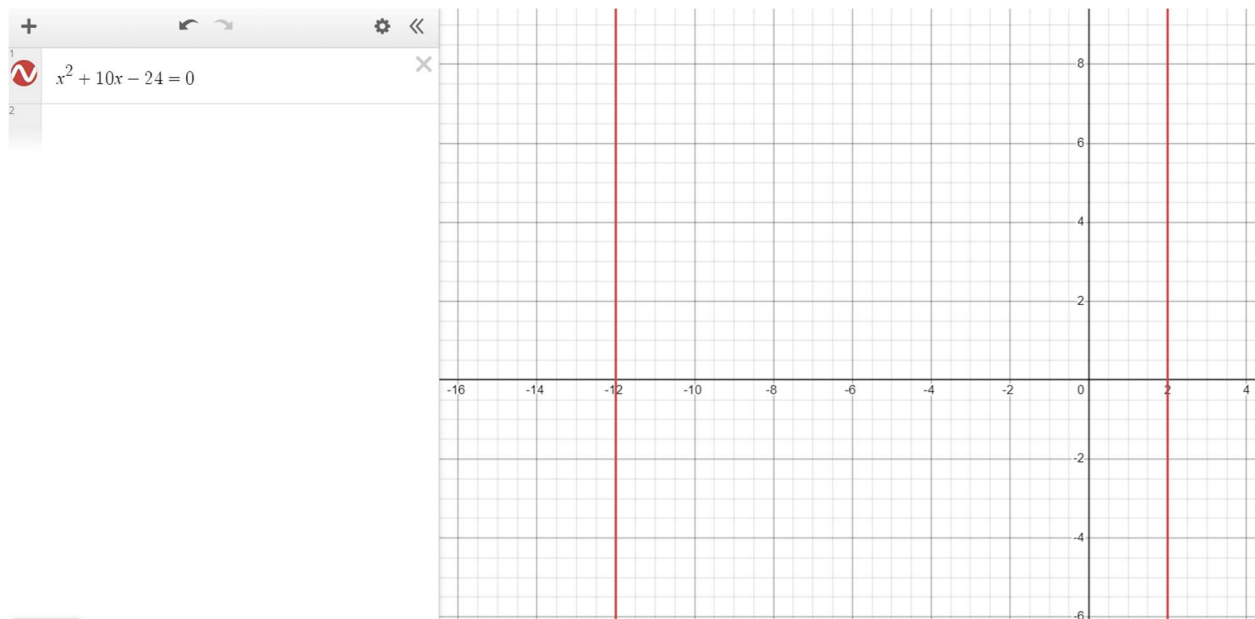
What is one of the solutions to the given equation?

Switch z to x and do the usual.

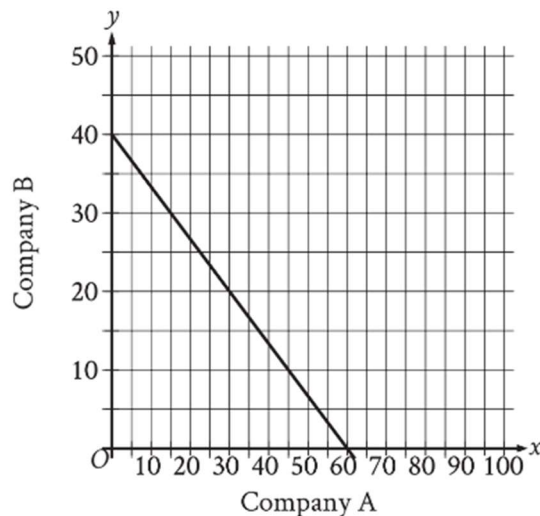
We can either do this (Remember, you can zoom in and out):



Or this (more convenient)



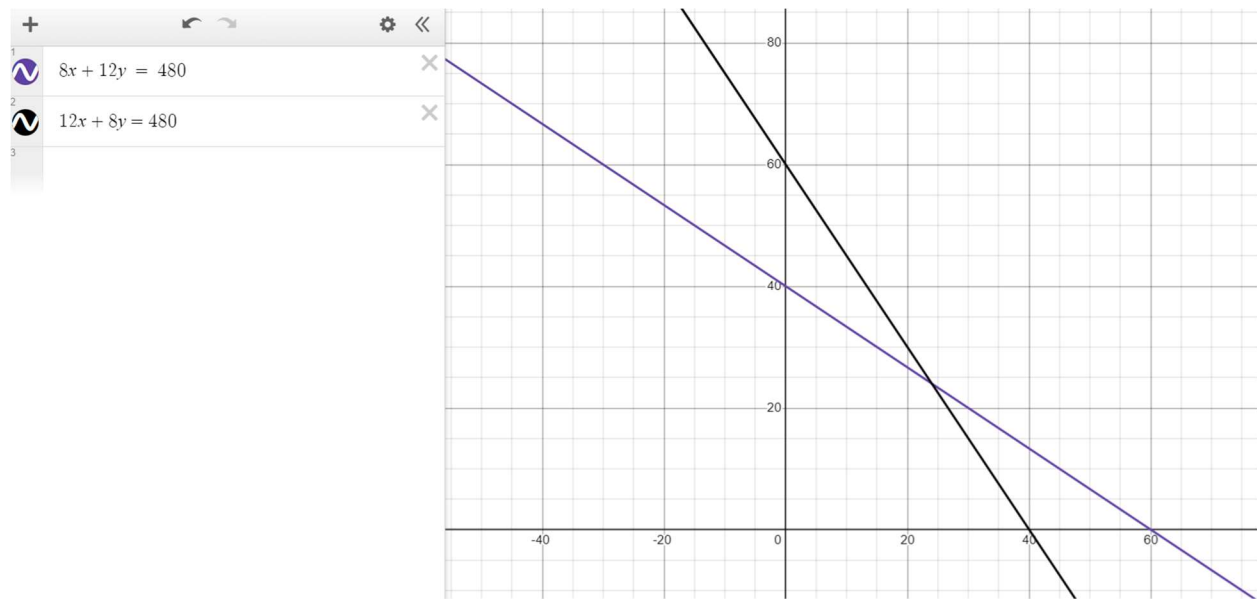
18



The graph shows the relationship between the number of shares of stock from Company A, x , and the number of shares of stock from Company B, y , that Simone can purchase. Which equation could represent this relationship?

- A) $y = 8x + 12$
- B) $8x + 12y = 480$
- C) $y = 12x + 8$
- D) $12x + 8y = 480$

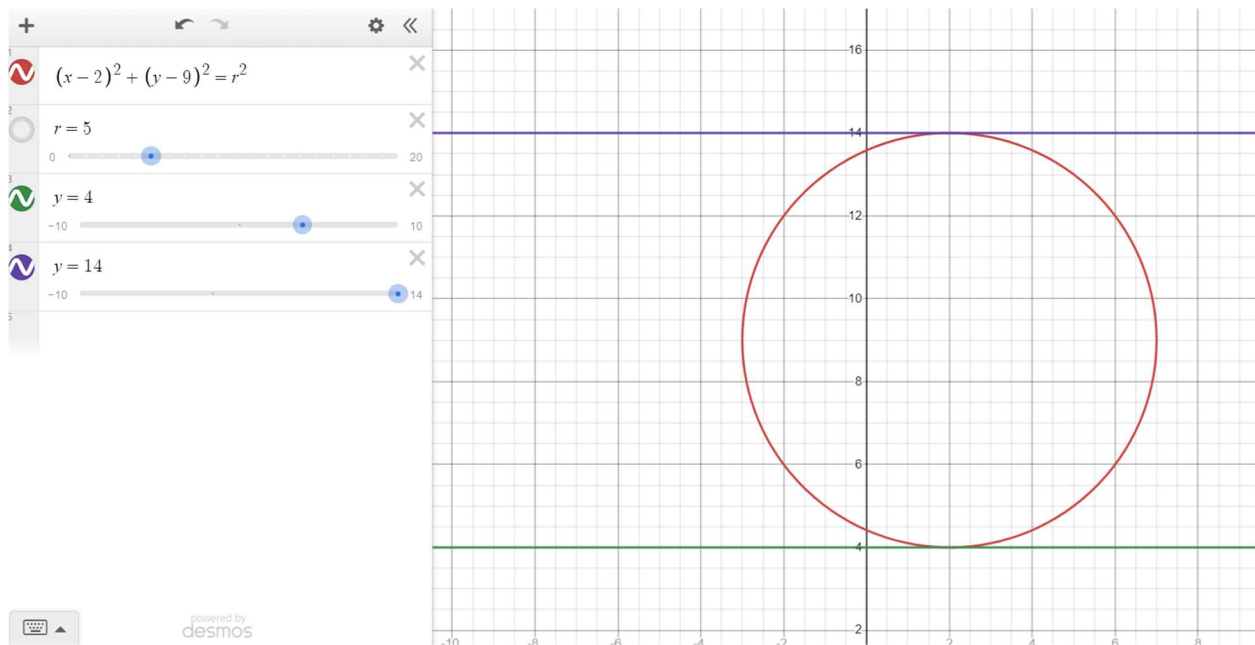
Using Desmos blindly for every question will hurt you. Instead, see if you can solve the question easily without Desmos or eliminate a few choices first. For example, here we can eliminate A and C because we notice the positive gradient but the line in question has a negative gradient.



21

A circle in the xy -plane has a diameter with endpoints $(2, 4)$ and $(2, 14)$. An equation of this circle is $(x - 2)^2 + (y - 9)^2 = r^2$, where r is a positive constant. What is the value of r ?

This is an interesting question to be solved using Desmos. As usual, we type out the equation as we see it, and add the slider for “ r ”. We then change the endpoints of r to be only within positive values (0 and 20), and we slide and see. We can also add 2 equations representing the diameter endpoints: $y = 4$ and $y = 14$.

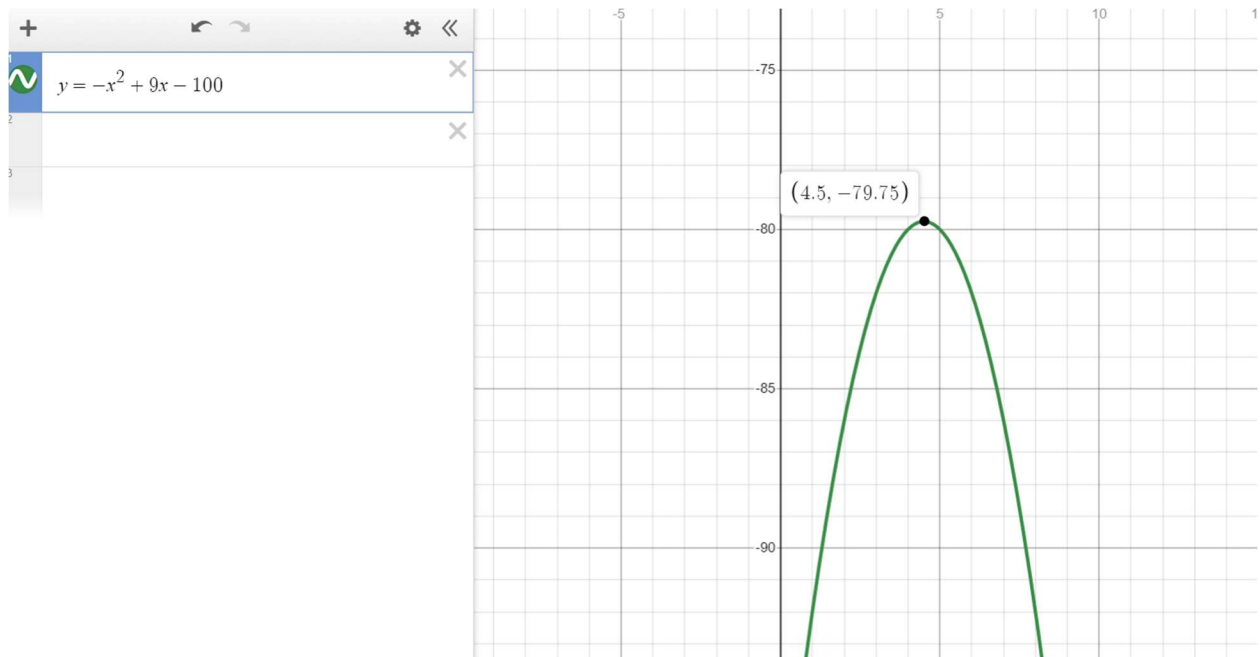


If it is too complex, no problem. You can just do the pen and paper calculations as taught in school. The shortest solution for the question would be to do $14 - 4$ and get diameter as 10, then divide by 2 to get $r = 5$.

25

In the xy -plane, the graph of the equation $y = -x^2 + 9x - 100$ intersects the line $y = c$ at exactly one point. What is the value of c ?

- A) $-\frac{481}{4}$
- B) -100
- C) $-\frac{319}{4}$
- D) $-\frac{9}{2}$



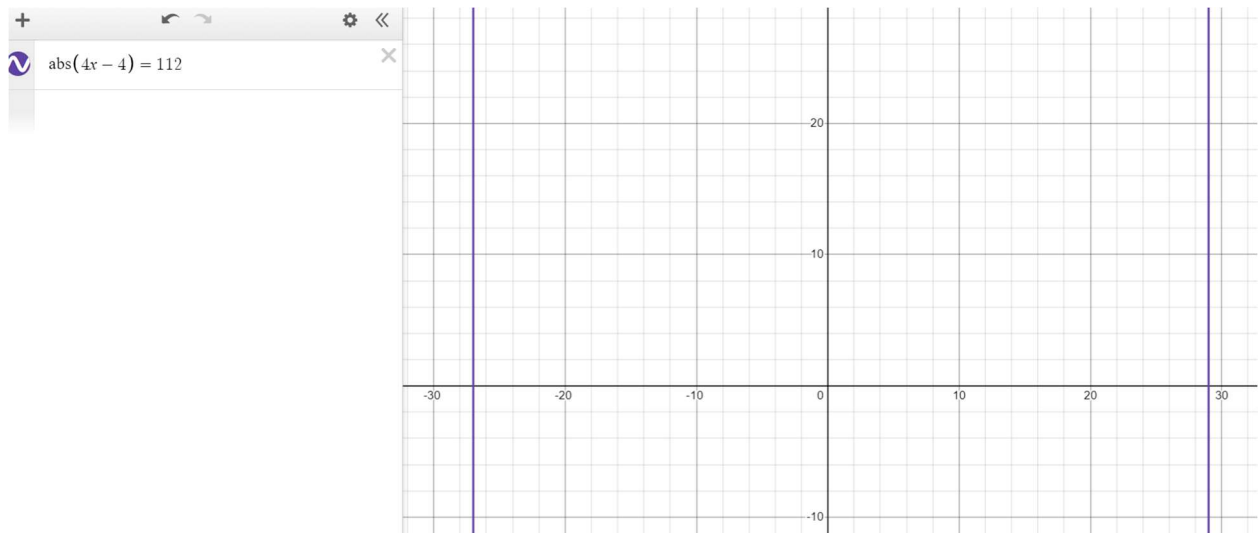
Then, if you need to get the answer, you can multiply the y-coordinate by 4 (answer is C)

$$-79.75 \times 4 = -319$$

21

If $|4x - 4| = 112$, what is the positive value of $x - 1$?

We can still use Desmos here:



We see that $x = -27$ and $x = 29$ are solutions. If you subtract 1, we get $x - 1 = -28$ or 28 . The positive value for both is the same, 28.

Final question:

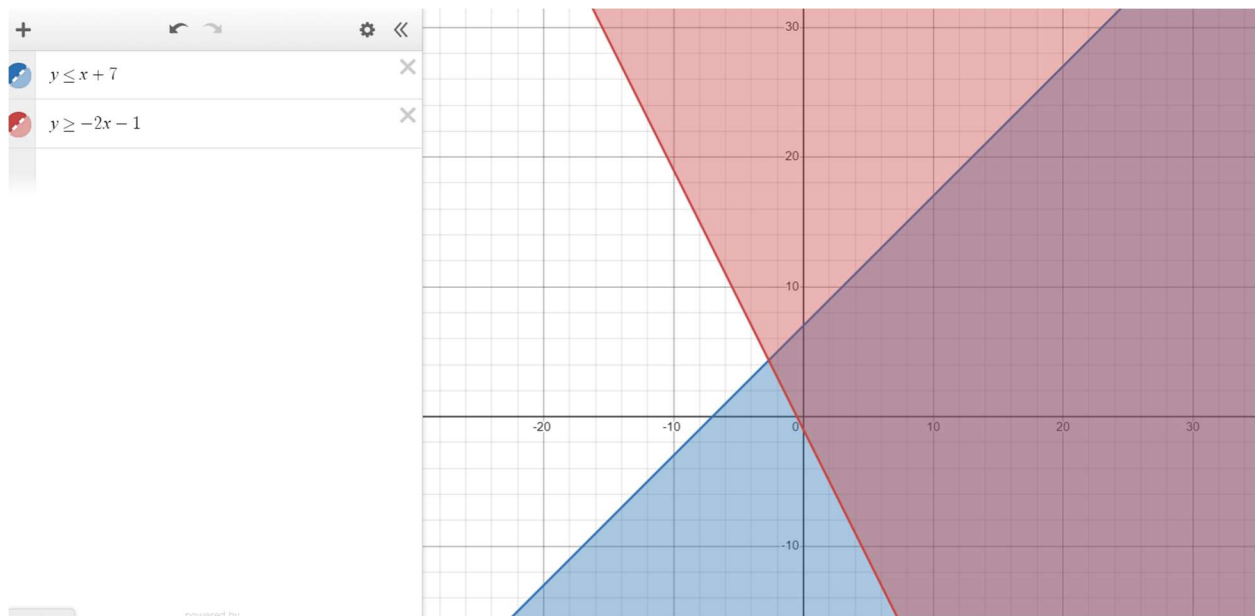
19

$$y \leq x + 7$$
$$y \geq -2x - 1$$

Which point (x, y) is a solution to the given system of inequalities in the xy -plane?

- A) $(-14, 0)$
- B) $(0, -14)$
- C) $(0, 14)$
- D) $(14, 0)$

We can also calculate inequalities in Desmos. Just use the appropriate symbols. If you want to show "less than or equal to", type "<=" then "="



The shaded regions are those that satisfy the inequality.

Additional Advice

Practice lots and lots of questions, and get familiar with Desmos. You will be able to judge when to use Desmos and when not to use it. However, by this point, you should know that Desmos is very powerful and can be applied to very many questions, even those that aren't graph-related.

Try and use Elimination where possible, as you might have seen in the earlier examples.

You have more than enough time to complete each math module. If you practice the different topics on Khan Academy, do full-length practice papers and extensively practice using the calculator, you should get a 750+.

Good luck!